



Research article

Robot path planning based on artificial potential field with deterministic annealing

Zhengtian Wu^{a,*}, Jinyu Dai^a, Baoping Jiang^a, Hamid Reza Karimi^{b,*}^a School of Electronic and Information Engineering, Suzhou University of Science and Technology, Suzhou, China^b Department of Mechanical Engineering, Politecnico di Milano, Milan, Italy

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ABSTRACT

In the context of motion planning in robotics, the problem of path planning based on artificial potential fields has been examined using different algorithms to avoid trapping in local minima. With this objective, this paper proposes a novel method based on a deterministic annealing strategy to improve the potential field function by introducing a temperature parameter to increase the robot's obstacle avoidance efficiency. The annealing and tempering strategies prevent the robot from being trapped at the local minima and allow it to continue towards its destination. The initial path is optimised using an annealing algorithm to enhance the overall performance. The time, length and success rate of the planned path measures the quality of the solution. Simulation results and comparative experiments demonstrate that the proposed algorithm can solve path planning in different environments. The proposed algorithm is suitable for complex environments with convex or non-convex polygon obstacles.

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1. Introduction

In robotics, the objective of path planning is to find the optimal path for a robot that satisfies the constraints according to a specific environment. With the development of intelligent control technology and artificial intelligence algorithms, the application of heuristic algorithms to solve the problem of robot path planning has become a research hotspot in the field of mobile robot control.

Path planning in partially or entirely unknown environments is a more realistic yet challenging task. The commonly used strategies include behaviour-based methods and intelligent control techniques, such as genetic algorithm (GA) [1,2], neural network algorithm [3,4], ant colony (ACO) algorithm [5,6] and particle swarm optimisation (PSO) algorithm [7,8].

Path planning in a known environment aims to generate the path to be followed by a robot, provided the whole environment, whether static or dynamic, is known. In this case, we can divide them into two broad categories: cost function minimisation algorithms and potential field-based methods. A typical example of the first algorithm is the heuristic A* algorithm [9,10]. Khatib [11] proposed the application of artificial potential field (APF) method for obstacle avoidance. Due to excellent planning speed and good

real-time performance of APF, it has outstanding advantages in path planning mainly because it provides simple, efficient and reliable motion planners for realistic targets. This method treats the robot as an object affected by an APF, and the target and obstacle are simulated by attractive and repulsive potentials, respectively. The sum of the potential functions above define the potential function of free space. Therefore, a generalised force equal to the negative gradient of the potential function is applied to the robot, driving it towards the target configuration. The APF can be used as a global and local method [12]. Indeed, this method yielded some work. For example, [13] reported a method inspired by deterministic annealing. When the robot is trapped in the local minima, the temperature parameter begins to increase until the robot escapes from the local minima. In [14], an ACO-APF algorithm was proposed to search for the global optimal path of an unmanned surface vehicle in the grid environment. To consider localisation accuracy in path planning, the method proposed in [15] used a mixture of potential and localisation risk fields to produce a hybrid directional flow to guide the safe and efficient path of an unmanned vehicle.

However, the APF has some inherent flaws in path planning [16], including (1) the absence of a feasible path in dense obstacle spaces; (2) the path trajectory goes beyond the equilibrium position, oscillating, or repeatedly closed-loop in the narrow space; and (3) it is trapped at the local minima before reaching the target. In order to address these problems, the following two issues are currently existed in the literature: (1) by redefining

* Corresponding authors.

E-mail addresses: wzht8@mail.usts.edu.cn (Z. Wu), dajinyu_stu@163.com (J. Dai), baopingji@163.com (B. Jiang), hamidreza.karimi@polimi.it (H.R. Karimi).

the potential function, removing or only keeping a small number of local minima (Local Minima Removal) and (2) avoiding obstacles (Local Minimum Avoidance) or escaping from the local minima (Local Minimum Escape) through intelligent search technology [17]. Aiming at the defects of unreachable target and the tendency to fall easily into the local minima in the standard potential field, various improvement methods exist in the literature. In [16], a path planning method that combines simulated annealing with APF, which provides the ability to escape any possible local minima, is introduced. In [18], when the target object is trapped in the local minima, the system generates a virtual obstacle to exert a repulsive force on the controlled object, thereby jumping out of the local minima. The authors in [19] combined APF with the ACO, which could enable the unmanned aerial vehicles to obtain a better feasible path. In [20], the local minima problem is solved by using the fuzzy model of obstacle repulsion and target gravity, which was used for the path planning of dynamic targets and obstacles. The authors in [21] introduced the control force to solve and improve the APF, to shorten the flight trajectory of the helicopter and smoothen the trajectory curve. In addition, in [22], an online deadlock avoidance method is proposed. By establishing a new repulsive potential field, the local minima problem is overcome. The authors in [23] proposed a method called ‘Dynamic Repulsive Potential Factor’ (DRPF). Compared with other existing improved combinatorial algorithms, such as sub-goal strategy and rapidly exploring random tree (RRT), DRPF has the advantage of simplicity and effectiveness.

Compared with [24], this paper focuses on the application of deterministic annealing strategy to path planning, using the feature that the change in temperature parameters during annealing and tempering fits the obstacle avoidance process of the robot, then incorporating and improving the classical APF algorithm, as well as using deterministic annealing for global optimisation. The authors in [24] focused on the algorithmic theoretical analysis of deterministic annealing, which has the advantage of solving the NP Hard problem, and completed the study of the maximum clique problem by constructing Lagrangian operators.

The main contribution of this paper is to propose a novel APF algorithm based on deterministic annealing (DA-APF). DA-APF is a modification of the standard APF method, where deterministic annealing strategies help the robot escape from the local minima, aided by a tempering strategy and can be used for local and global path planning. The DA-APF algorithm is proposed in detail by taking the path planning of a model robot in a 2D convex polygon or non-convex polygon obstacle environment. In this work, the problem is simplified to make it suitable for the study of annealing parameters, but the method can be extended to more complex environments.

The remainder of this paper is organised as follows: Section 2 introduces the basic idea of the DA method. Section 3 defines the improved potential function with the introduction of the temperature parameter and then presents the DA-APF algorithm. In Section 4 presents some simulation results, which demonstrate the effectiveness of the proposed algorithm. Finally, Section 5 summarises this paper.

2. Deterministic annealing

Deterministic annealing [25] is a continuous global optimisation technique with temperature as the homotopy parameter that can solve the global minimum of non-convex problems. Referring to the thermodynamic annealing process, the temperature parameter is introduced into the objective function, and the optimal value of the objective function is transformed into the minima of a series of free energy functions that vary with temperature. It can avoid many local minima that have perplexed the traditional technology without the slow scheduling usually required

by the stochastic method. The solution obtained by this technique is completely independent of the selection of the initial configuration and yields a non-stochastic solution under the limit of ‘temperature’ 0. DA has been widely used in the travelling salesman problem [26,27], which can efficiently obtain the optimal path. Some scholars also apply DA to vehicle path planning problems [28,29].

To solve minimisation problems: $\min E = E(x)$, where $E(x)$ is the energy of a certain system. When solving this problem with DA, the optimal solution can be obtained from the minima value of the system’s free energy function. A corresponding free energy function of the system [29] is constructed: $F(x, T) = E(x) - TH$, where T and H are the temperature and entropy of the system, respectively. This function is solved according to the following process:

1. Take a sufficiently large $T_0 (T_0 = \infty)$, $k = 0$, and solve the optimisation problem: $\min F(x, T_k)$, let the optimal solution be $x_{\min}(T_k)$.
2. $T_{k+1} = cT_k (0 < c < 1)$, take $x_{\min}(T_k)$ as the initial solution, select an iterative method (such as Newton iteration method, conjugate gradient method or gradient descent method) to solve $\min F(x, T_{k+1})$, and let the optimal solution be $x_{\min}(T_{k+1})$.
3. Judge whether the convergence criterion is met. If so, then Stop, $x_{\min}(T_{k+1})$ is the optimal solution. Otherwise, turn to (4).
4. $k = k + 1$, turn to (2).

As shown in Fig. 1, the energy function is controlled by the temperature parameter, and as T approaches infinity, the free energy function is usually a convex function whose global minimum is easily identified. Therefore, DA can obtain the global minimum of the free energy function at infinite temperature. With the gradual decrease in temperature, this value continuously tends to move towards the global minimum of the system energy function and reaches this value when the temperature is equal to zero.

Ref. [30] proved that the initial point of this technique should be chosen at high temperatures because the deinitialisation at higher temperatures could alleviate nonlinear effects. The entropy at high temperature is maximised, gradually reducing this constraint as the temperature decreases, and directly minimising the energy function of the system when the temperature is zero, thereby resulting in a non-stochastic solution. The cooling schedule has a crucial effect on the performance of the technology. Rapid cooling results in quenching effect, which is likely to lead to the non-global minimum at zero temperature. Slow cooling can keep the system in equilibrium. Thus, at zero temperature, the energy configuration of the system is the smallest, and the best solution can be obtained. DA allows the geometric cooling law: $T(t) = \rho T(t-1)$, $0 < \rho < 1$, and ρ usually takes between 0.95 and 0.99. Ref. [31] showed that the probability convergence of the global minimum could be achieved when $T(t) = O((\log t)^{-1})$, where $T(t)$ is the temperature at time t .

3. APF with deterministic annealing

In this section, an improved APF method with deterministic annealing is proposed. It begins with the discussion of standard APF methods in Section 3.1. Then, the gradient descent potential-guided strategy and the local minima problem are introduced in Section 3.2. Next, the improved potential function is proposed, and the DA-APF algorithm is introduced in Section 3.3.

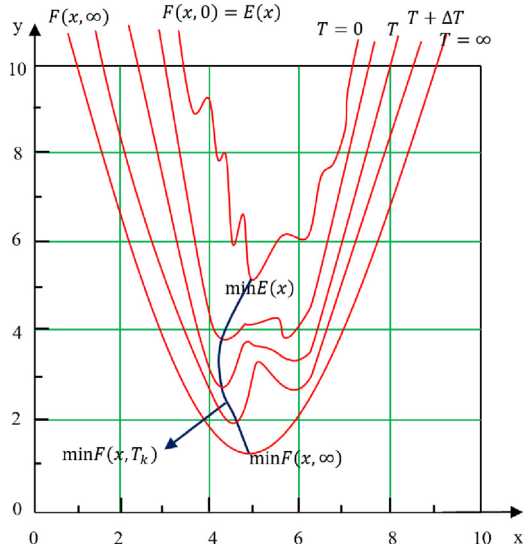


Fig. 1. Solution idea of deterministic annealing technology.

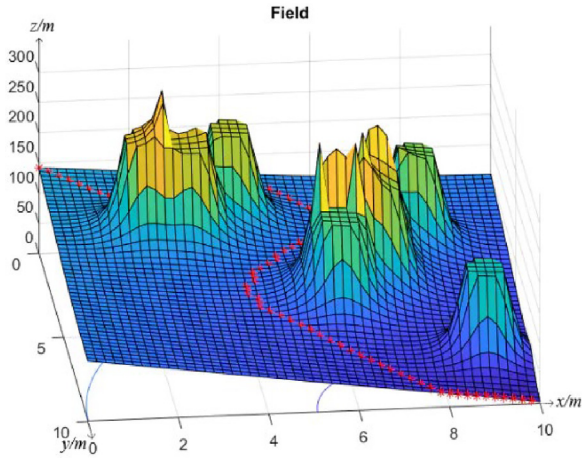


Fig. 2. Path planning in APF.

3.1. Potential field function

In this section, we will introduce the APF method and develop the appropriate equations required to ensure the safe navigation of the robot. By imitating the electric potential field, APF calculates the potential energy of each position according to the obstacles and targets in the scene and then allows the robot to move forward along the negative gradient direction (the direction of potential energy decline) to reach the target (Fig. 2), the peak convex hull represents the obstacle and its vicinity, the red dotted line represents the moving path of the robot, and the endpoint of the path has the lowest potential energy.

Figs. 3 and 4 show that the APF includes an attractive field and a repulsive field, in which the target exerts the attractive force on the object, guiding the object to move towards it. The obstacles generate the repulsive force on the object to avoid collision between the object and it. The resultant force on an object at each point on the path is equal to the sum of all repulsive and attractive forces at this point. The key is how to construct the attractive and the repulsive fields, which will be introduced separately below. In this paper, the distances at five specific angles are measured to calculate the repulsive force,

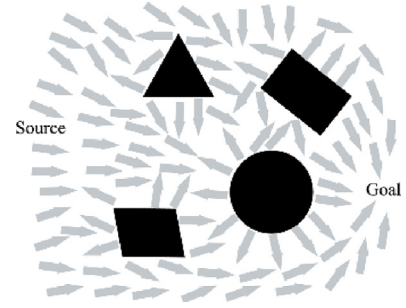


Fig. 3. Attractive and repulsive force.

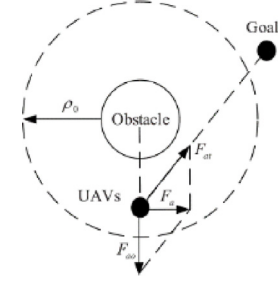


Fig. 4. Force analysis.

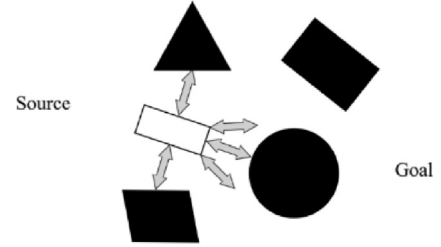


Fig. 5. Measurement of repulsive force.

which are forward, left, right, forward left diagonal and forward right diagonal as shown in Fig. 5.

3.1.1. Attractive potential function

The attractive field can attract the robot to the target. The basic idea is that the attractive field increases gradually as the robot moves away from the target. The attractive potential function can take many forms, but the most commonly used are conic and quadratic forms [11]. Considering the zero point in the calculation method is necessary because the conic function has the problem of discontinuous target position. Therefore, in this paper, the quadratic potential is described as follows [15]:

$$U_{att}(q) = \begin{cases} \frac{1}{2} \xi \sigma^2(q, q_{target}), & \sigma(q, q_{target}) \leq \sigma_0 \\ \xi \sigma(q, q_{target}) \sigma_0 - \frac{1}{2} \xi \sigma_0^2, & \sigma(q, q_{target}) > \sigma_0 \end{cases}, \quad (1)$$

where $U_{att}(q)$ is the numerical value of the attractive field; and ξ is the attractive gain constant, which is a positive parameter. $q_{target} = (x_{target}, y_{target})^T$ is the position of the target, $q = (x, y)^T$ is the current position of the robot, and $\sigma(q, q_{target}) = \|q - q_{target}\|_2$ represents the Euclidean distance between the robot and the target. σ_0 is the distance threshold between the robot position and the target position. Therefore, considering the position of the robot, the magnitude of the attractive force generated by

the target is defined as the negative gradient of the attractive potential function:

$$\vec{F}_{att}(q) = -\nabla U_{att}(q) = \begin{cases} -\xi(q - q_{target}), & \sigma(q, q_{target}) \leq \sigma_0 \\ -\frac{\xi\sigma_0(q - q_{target})}{\sigma(q, q_{target})}, & \sigma(q, q_{target}) > \sigma_0 \end{cases} \quad (2)$$

In the design of this attractive field, the distance threshold σ_0 between the robot and the target is considered. When the current distance is less than σ_0 , the farther the distance is, the larger the gradient will be, and the greater the attractive force the robot receives. When the current distance is greater than σ_0 , the farther the distance is, the smaller the gradient will be, and the less the attractive force the robot receives, which can avoid the problem of collision with obstacles caused by too much attractive force because of the large distance.

Correspondingly, the components in both directions can be described as follows:

$$\vec{F}_{att_x}(q) = \begin{cases} -\xi(x - x_{target}), & \sigma(q, q_{target}) \leq \sigma_0 \\ -\frac{\xi\sigma_0(x - x_{target})}{\sigma(q, q_{target})}, & \sigma(q, q_{target}) > \sigma_0 \end{cases} \quad (3)$$

$$\vec{F}_{att_y}(q) = \begin{cases} -\xi(y - y_{target}), & \sigma(q, q_{target}) \leq \sigma_0 \\ -\frac{\xi\sigma_0(y - y_{target})}{\sigma(q, q_{target})}, & \sigma(q, q_{target}) > \sigma_0 \end{cases} \quad (4)$$

where $\vec{F}_{att_x}(q)$ is the attractive force along the x-axis and $\vec{F}_{att_y}(q)$ is the attractive force along the y-axis.

3.1.2. Repulsive potential function

The repulsive field generates the repulsive force that keeps the robot away from obstacles. However, if the robot and the obstacle have a certain distance threshold, then its movement must be guaranteed not to be affected in any way. The repulsive function [15] is given by

$$U_{repi}(q) = \begin{cases} \frac{1}{2}\eta(\frac{1}{\rho_i(q, q_{obst})} - \frac{1}{\rho_0})^2, & \rho_i(q, q_{obst}) \leq \rho_0 \\ 0, & \rho_i(q, q_{obst}) > \rho_0 \end{cases} \quad (5)$$

If the scene contains n obstacles, the total repulsive potential field is expressed as follows:

$$U_{rep}(q) = \sum_{i=1}^n U_{repi}(q), \quad (6)$$

where $U_{repi}(q)$ is the value of the repulsive field for each obstacle i , and η is the repulsion gain constant, which is a positive parameter. $q_{obst} = (x_{obst}, y_{obst})^T$ indicates the position closest to the obstacle, and $q = (x, y)^T$ is the current position of the robot, $\rho_i(q, q_{obst}) = \|q - q_{obst}\|_2$ represents the distance from the robot to the nearest point of each obstacle. ρ_0 is the radius of influence of the obstacle. Therefore, the magnitude of the repulsive force generated by the obstacle is the negative gradient of the repulsive potential function:

$$\vec{F}_{repi}(q) = -\nabla U_{repi}(q) = \begin{cases} \eta(\frac{1}{\rho_i(q, q_{obst})} - \frac{1}{\rho_0})\frac{1}{\rho_i^2(q, q_{obst})}\frac{q - q_{obst}}{\rho_i(q, q_{obst})}, & \rho_i(q, q_{obst}) \leq \rho_0 \\ 0, & \rho_i(q, q_{obst}) > \rho_0 \end{cases} \quad (7)$$

The distance threshold ρ_0 between the robot and the obstacle is considered in the design of this repulsive field. The closer the current position is to the obstacle, the larger the gradient and the greater the repulsive force the robot receives. When the current

position and the obstacle are greater than ρ_0 , the robot is no longer repulsed by the obstacle.

The expressions for the Cartesian components $\vec{F}_{repi_x}(q)$ and $\vec{F}_{repi_y}(q)$ of the repulsive force can be written as follows:

$$\vec{F}_{repi_x}(q) = \begin{cases} \eta(\frac{1}{\rho_i(q, q_{obst})} - \frac{1}{\rho_0})\frac{1}{\rho_i^2(q, q_{obst})}\frac{x - x_{obst}}{\rho_i(q, q_{obst})}, & \rho_i(q, q_{obst}) \leq \rho_0 \\ 0, & \rho_i(q, q_{obst}) > \rho_0 \end{cases} \quad (8)$$

$$\vec{F}_{repi_y}(q) = \begin{cases} \eta(\frac{1}{\rho_i(q, q_{obst})} - \frac{1}{\rho_0})\frac{1}{\rho_i^2(q, q_{obst})}\frac{y - y_{obst}}{\rho_i(q, q_{obst})}, & \rho_i(q, q_{obst}) \leq \rho_0 \\ 0, & \rho_i(q, q_{obst}) > \rho_0 \end{cases} \quad (9)$$

3.1.3. Total potential function

After calculating the attractive potential function and the repulsive potential function respectively, the total potential function is obtained by summing them together:

$$U(q) = U_{att}(q) + \sum_{i=1}^n U_{repi}(q). \quad (10)$$

Therefore, the corresponding resultant force is expressed as follows:

$$\vec{F}(q) = \vec{F}_{att}(q) + \sum_{i=1}^n \vec{F}_{repi}(q). \quad (11)$$

3.2. Gradient descent potential guided strategy and local minima

As mentioned in the previous section, the attractive and repulsive potential functions can be developed independently, and the total potential function is the sum of these potential energies, which is usually used to control the robot to move along the predetermined trajectory whilst avoiding obstacles to achieve the target. Therefore, the trajectory of the robot is obtained by tracking the resultant force generated by the sum of attractive and repulsive forces. This resultant force is calculated as the negative gradient of the total potential function. This method is called the gradient descent potential guided strategy.

According to the obtained APF, start from the source, calculate the gradient of the current position, advance in the negative direction of the gradient, then calculate the gradient of the current position again, and proceed in the negative direction of the gradient until the target is reached. This approach includes the calculation of the position of the robot at each step in the direction of the resultant force. Before calculating the resultant force, the attractive force, the Euclidean distance between the robot and the nearest obstacle, and the final repulsive force must be calculated. The pathfinding process through the gradient descent potential guided strategy is presented as follows:

1. Set $q_0 = q_{source}$, $i = 0$.
2. If $q_i \neq q_{target}$, then calculate the attractive force $\vec{F}_{att}(q)$.
3. Calculate the real-time distance $\rho_i(q, q_{obst})$ between the robot and the nearest obstacle, and thus calculate $\vec{F}_{repi}(q)$.
4. Calculate the resultant force $\vec{F}(q)$.
5. Set $q_{i+1} = q_i + step \times \frac{\vec{F}(q)}{\|\vec{F}(q)\|_2}$, if $q_{i+1} = q_{target}$, Stop. Otherwise, turn to (6).
6. $i = i + 1$, turn to (2).

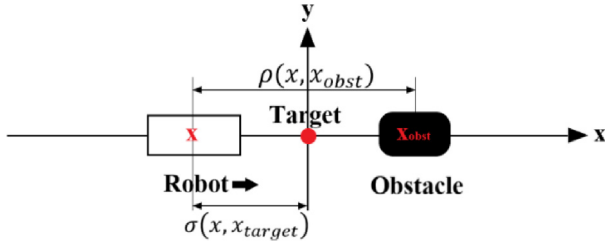


Fig. 6. Local minima in a one-dimensional environment.

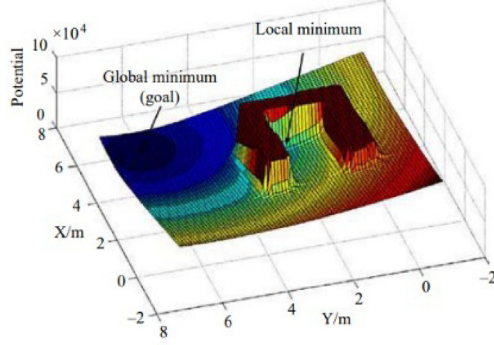


Fig. 7. U-shaped obstacle environment.

However, the gradient descent in optimisation theory tends to be stuck in the local optimum [32]. Similarly, gradient descent has this disadvantage in path planning. This paper provides a simple example in a one-dimensional environment. The current position of the robot is collinear with the obstacle and the target (Fig. 6). The use of APF in such an environment often falls into the local minima, making it difficult to reach the target. The following is explained from a mathematical point of view.

To simplify the description above, making the robot move only on the x-axis is advisable. The path source is $q_0 = (x, 0)^T$, the target is $q_{target} = (0, 0)^T$, and an obstacle is placed at $q_{obst} = (x_{obst}, 0)^T$. Assuming that both the obstacle and the target are within the limit distance threshold of potential field influence, the Eqs. (1) and (5) can be rewritten as follows:

$$U_{att}(x) = \frac{1}{2}\xi x^2, \quad (12)$$

$$U_{rep}(x) = \frac{1}{2}\eta \left(\frac{1}{x - x_{obst}} - \frac{1}{\rho_0} \right)^2. \quad (13)$$

Let us set $x_{obst} = 1$, $\rho_0 = 2$, $\xi = \eta = 1$, then ones read

$$\begin{aligned} U(x) &= U_{att}(x) + U_{rep}(x) \\ &= \frac{1}{2}x^2 - \frac{1}{2} \left(\frac{1}{x-1} - \frac{1}{2} \right)^2. \end{aligned} \quad (14)$$

It is not difficult to find from formula (14) that when $x = 0$, that is, when the robot reaches the predetermined target, the total potential function at this time does not reach the minimum. The robot will be stuck at $x = -1$, that is, the resultant force $F(q) = 0$, and thus, it can no longer move forward. It can be expected that the robot will sway left and right in a straight line, failing to reach the target. At this time, it is necessary to give the robot a disturbance to make its escape from the local minimum.

The three aforementioned points and one line are simple trap environment. In particular, this paper is aimed at the classical trap environment of the local minimum, that is, U-shaped obstacles, as shown in Fig. 7.

3.3. APF with deterministic annealing

In this section, facing the local minima is still an important research content in the APF method, and a new algorithm based on deterministic annealing strategy and optimisation theory, DA-APF, is proposed.

The method of escaping from the local minima includes random searches on the surface of the potential energy function and attempting to minimise the potential energy function to find a position with lower potential energy than the local minima where the current robot is stuck. Inspired by the deterministic annealing process, the DA-APF introduces the temperature parameter to control the trajectory of the robot, which can effectively avoid its random movement on the surface of the potential energy function. Reconsider Formulas (1) and (5):

$$U_{t-att}(q) = \frac{1}{2}\xi \frac{\sigma^2(q, q_{target})}{\alpha T}, \quad \frac{\sigma(q, q_{target})}{T} \leq \sigma_0 \quad (15)$$

$$U_{t-repi}(q) = \begin{cases} \frac{1}{2}\eta \left(\frac{1}{\frac{\rho_i(q, q_{obst})}{\alpha T}} - \frac{1}{\rho_0} \right)^2, & \frac{\rho_i(q, q_{obst})}{T} \leq \rho_0 \\ 0, & \frac{\rho_i(q, q_{obst})}{T} > \rho_0 \end{cases} \quad (16)$$

Then, we have

$$U_t(q) = U_{t-att}(q) + \sum_{i=1}^n U_{t-repi}(q), \quad (17)$$

where T is the temperature parameter, α is the cooling factor, and $0 < \alpha < 1$. Define $H = -\sum_q p(q) \log p(q)$, $p(q)$ represents the probability of the robot passing through q , and the following Lagrangian-type free energy potential function that corresponds to the system is constructed:

$$E(q, T) = U_t(q) - TH. \quad (18)$$

Formulas (15) and (16) are potential field functions that are improved by introducing the temperature parameter, which can reduce the distance threshold σ_0 between the robot and the target and increase the influence radius ρ_0 of obstacles. If $T = 1$ is used as the initial state, then Formulas (17) and (10) are equivalent, and the newly constructed potential field has the same initial path selection as the original potential field. When the robot is trapped in the local minima, the temperature begins to increase until the robot moves out of the influence range. Then, the temperature begins to drop slowly to 1, at which time the attractive and the repulsive potential fields return to the states of Formulas (1) and (5). However, the aforementioned strategies may produce inefficient paths because the initial state is not optimal. In the early stage, the robot still performs path planning based on reaching the target, and the annealing strategy will be executed only when the local minima are captured. In other words, this is an escape plan for local minima.

The DA-APF is further improved to improve the overall planning performance. Now, the algorithm will initialise at a sufficiently high temperature, which theoretically conforms to the basic idea of deterministic annealing, that is, in the initial high-temperature state, the potential energy functions (15) and (16) are usually convex, and their global minimum is easy to obtain. With the annealing process, this global minimum is tracked to the local minima of the original potential function and is defined the temperature as 1 at this time.

From an intuitive point of view that the robot will be more sensitive to obstacles, that is, in the early stage of path planning, when the temperature is high, the robot will focus on the search and avoidance of obstacles rather than blindly running to the

target. In the later planning stage, with the slow decline of temperature, the repulsive force decreases, and the attractive force increases, the robot appears to gradually lose ‘interest’ in obstacles and rush to the target with all its strength. The tempering strategy will be adopted if the robot falls into the local minima during the planning process. Tempering is the inverse process of annealing, equivalent to returning to the last moment of the iterative process, inheriting the existing intermediate variables, and is, therefore, very efficient. For this reason, the annealing and tempering processes form a systematic regulation mechanism. At this point, the temperature will stop falling and instead increase. The repulsive force of the obstacle to the robot will increase sharply to ‘pop’ the robot out of the local minima. This process is called the obstacle avoidance method. The robot can achieve fast pathfinding and efficient obstacle avoidance by increasing the potential influence of obstacles and increasing the priority of the local minima avoidance in path planning.

Based on the deterministic annealing strategy, the DA-APF for global path planning includes the following modules:

A. Path planning main programme

1. For the potential field pre-processing, use deterministic annealing to obtain the global minimum of the potential field function $U_t(q)$, denoted as p .
2. Set P_0 as the initial path from the source to the target, ignoring the existence of obstacles.
3. Set $P_{min} = P_0$, where P_{min} is the shortest path without passing through p and use the gradient descent potential guided strategy is used for planning.
4. When P_{min} collides or falls into the local minima, repeat the following steps:
 - 4.1. Set $P = P_{min}$, and use deterministic annealing to help escape.
 - 4.2. If the escape succeeds, then let $P_{min} = P$, and continue planning with gradient descent potential guided strategy.
5. If there is no Failure, then return to P_{min} .

B. Potential field pre-processing (Initial path optimisation)

1. Set $T_0 = \infty$, $k = 0$, solve $\min E(q, T_k)$, and let the optimal solution be $p(T_k)$.
2. Set $T_{k+1} = \alpha T_k$ ($0 < \alpha < 1$), take $p(T_k)$ as the initial solution, use the conjugate gradient method to solve $\min E(q, T_{k+1})$, and let the optimal solution be $p(T_{k+1})$.
3. If $|p(T_{k+1}) - p(T_k)| \leq \text{tolerance}$, stop annealing and let $p(T_{k+1})$ be the optimal solution, which is denoted as the global minimum p of $U_t(q)$. Otherwise, turn to (4).
4. $k = k + 1$, turn to (2).

C. Gradient descent potential guided strategy

1. Set $P = P_0$.
2. Repeat the following steps until all points in P are traversed:
 - 2.1. Let q be the point to be traversed with the highest potential energy $U_t(q)$ in P .
 - 2.2. Calculate the potential $U_t(\tilde{P}_i)$ of all paths that passing through q and within the neighbourhood of P .
 - 2.3. Let $\tilde{P}$ be the path with the lowest potential energy in the above neighbourhood.
 - 2.4. If $U_t(\tilde{P}) \leq U_t(P)$, then let $P = \tilde{P}$. Otherwise, the point q has been traversed.
3. Return to P .

D. Deterministic annealing

1. Set $P = P_0$, $T = T_0$, where T_0 is large enough.
2. Start annealing, set $T_{i+1} = \alpha T_i$ ($0 < \alpha < 1$), and use the conjugate gradient method to obtain the p_i that corresponds to each T_i .
3. When $T_i \geq T_f$, repeat the following steps:
 - 3.1. Under each T_i , if p_i is not on P , then execute the gradient descent potential guided strategy to obtain $\tilde{P}$. Let $P = \tilde{P}$.
 - 3.2. If p_i is in the neighbourhood of P under a certain T_i , then execute the loop:
 - 3.2.1. If $d(p_i, P) \leq \rho_0$, then execute tempering strategy. Let $T_{i+1} = \frac{1}{\alpha} T_i$ until escape, and then execute the gradient descent potential guided strategy to obtain $\tilde{P}$. Let $P = \tilde{P}$.
 - 3.2.2. If $d(p_i, P) > \rho_0$, then repeat Step 3.1.
 - 3.3. If $U_t(\tilde{P}) < U_t(P_0)$, then obtain the optimal path P_{min} . Let $P = P_{min}$.
 - 3.4. Let $T_{i+1} = \alpha T_i$.
4. If there is a collision or no escape, then return to Failure. Otherwise, return P .

To sum up, the DA-APF combines the real-time performance of the APF and integrates the ability of DA to solve the global minimum. Before planning the path, the global minimum on the surface of the improved potential energy function is obtained by DA, which can optimise the initial path to the greatest extent. Then, according to the trend of the potential field, path planning is guided by gradient descent. Suppose that the process falls into the local minima. In that case, the tempering strategy is used to help escape, and then annealing is continued until the obstacle avoidance is completed or the target is reached.

4. Simulation results

This section introduces three experiments to demonstrate the effectiveness of the proposed DA-APF. The comparative experiments are designed to illustrate that the proposed algorithm performs well in various specific environments.

4.1. Simulation configuration

Our simulation studies are conducted in MATLAB (R2016a) under Windows 10 on a laptop with a 2.50 GHz AMD Ryzen 5 4600U CPU and 16.0 GB RAM. The experiment is conducted on the basis of the simulator developed by MATLAB. The simulated robot is a differential driven mobile platform equipped with five distance sensors that are evenly distributed on the front, left, right, left front and right front sides of the robot, and the detection distance is about 3 m. This paper considers the Euclidean distance and acceleration to accelerate the calculation of the closest distance of the obstacle to the robot centre (as a control point). All 2D environments contain the local minima that may prevent the robot from reaching the target. In the figures that represent the simulation results, the source is represented by red or S, and the target is represented by green or T. The distance units of the coordinate axis are all decimetre. For all experiments, the parameters are set as follows: $\xi = 1$, $\eta = 3$, $\sigma_0 = 100$, $\rho_0 = 30$, $T_0 = 10\,000$, and $\alpha = 0.98$.

The three designed experiments include the simulation of DA-APF in ‘trap’ environments, the simulation of DA-APF in stochastic

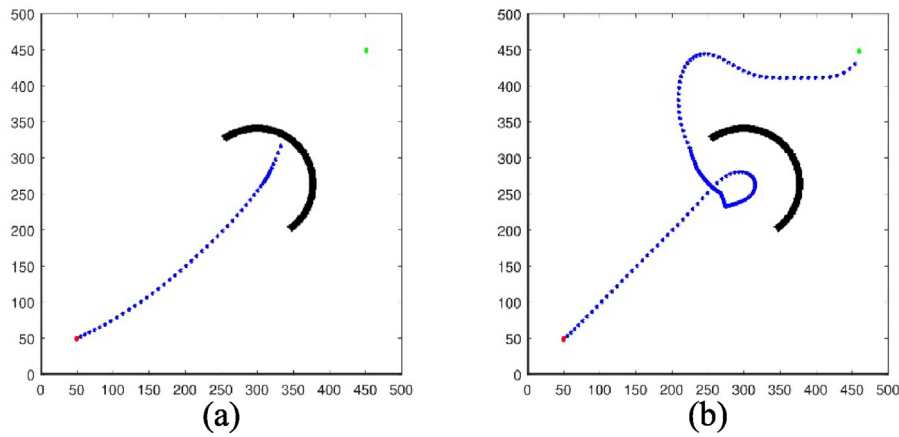


Fig. 8. Environment One ((a) and (b): standard APF, DA-APF).

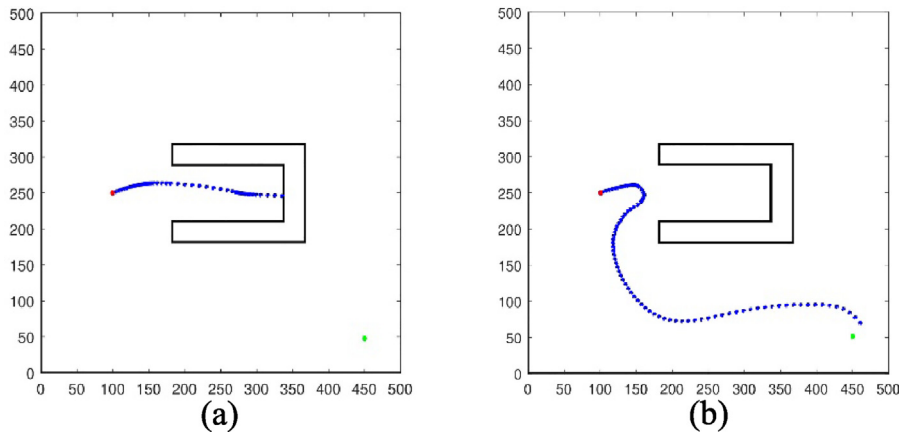


Fig. 9. Environment Two ((a) and (b): standard APF, DA-APF).

simple multi-objective environments and the simulation of DA-APF and other path planning algorithms in complex environments. Amongst them, the standard APF is used for comparative simulation in the first experiment, and the bidirectional rapidly exploring random tree algorithm (B-RRT) [33], Probabilistic roadmap (PRM) algorithm [34], GA [35] and improved A-star (IA*) algorithm [36] are used for comparative simulation in the third experiment. To describe the performance of the DA-APF from multiple perspectives comprehensively, the path length, pathfinding time, path smoothness and pathfinding success rate are used as evaluation indicators.

4.2. Experiment one

This section is simulated in the trap environment or the scene of U-shaped obstacles, which is a classic configuration of the local minima. The dimensions of the environment are all 500 dm $\times$ 500 dm.

4.2.1. Environment One

The first simulated experiment is the process of obstacle avoidance through a simple U-shaped obstacle. The coordinates of the source and the target are (50, 50) and (450, 450), respectively. As shown in Fig. 8(a), when using the standard APF without any obstacle avoidance mechanism, the robot will fall into the local minima within the U-shaped obstacle and cannot escape. However, the DA-APF can guide the robot to the target and

complete obstacle avoidance. The path in Fig. 8(b) shows that the robot has two guiding difficulties, namely the turning process when entering the influence distance of the obstacle and the continuing pathfinding process after escaping from the influence range.

4.2.2. Environment Two

The second environment is also a U-shaped obstacle. The difference from the first environment is that the size of the simulated obstacle is larger, and the local minima are very strong. The coordinates of the source and the target are (100, 250) and (450, 50), respectively. Fig. 9(a) shows that it is difficult for the standard APF to complete obstacle avoidance, and the robot is trapped in the U-shaped obstacle. The proposed DA-APF can solve the local minima problem, as shown in Fig. 9(b), the robot completes the path planning without entering the U-shaped obstacle, the obtained path is relatively smooth, and only the initial selection process fluctuates. At the same time, the temperature initialisation of the DA-APF will affect the path length. When the initial temperature is too high, the robot will be too far away from the obstacles.

4.2.3. Environment Three

Different from Environments One and Two, the setting of Environment Three considers the influence of temperature on the robot's pathfinding process. The coordinates of the source and the

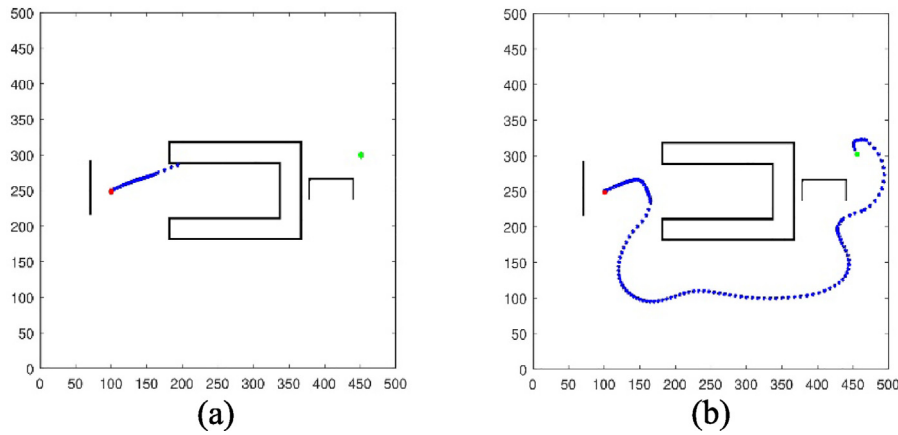


Fig. 10. Environment Three ((a) and (b): standard APF, DA-APF).

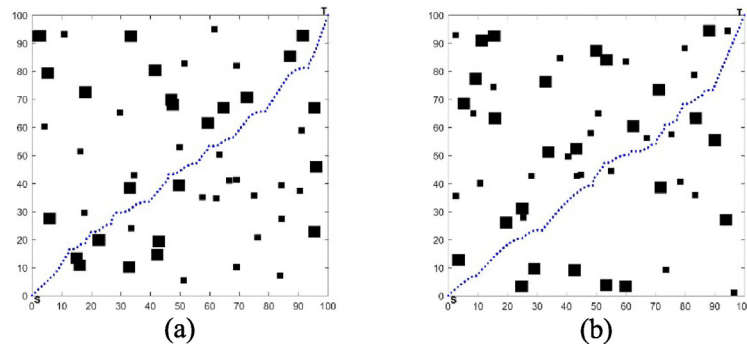


Fig. 11. Stochastic simple multi-objective environment ((a): step-size = 1.5, (b): step-size = 0.75).

target are (100, 250) and (450, 300), respectively. For the DA-APF, the unknown obstacles will not act on the potential field because the robot cannot detect the information of all obstacles in the early planning stage. If the initial temperature is too low, the robot may fall into the local minimum inside an obstacle because the temperature has already been sufficiently low when the robot is about to reach the target. At this time, the repulsive force of the obstacle is already very small, and the robot will no longer perform obstacle avoidance operations. Because the simulated environment allows the existence of the local minima, low temperature at this time will create the illusion that there are no obstacles for the robot, resulting in failure of obstacle avoidance. The solution is to temper and raise the temperature again. Once it escapes from the local minimum, it cools down slowly until it reaches the target.

Fig. 10(a) shows that the standard APF cannot guide the robot to reach the target, but the DA-APF completes the planning, as shown in Fig. 10(b). Because the DA-APF is initialised at a high temperature, the robot will have a trajectory away from the target whilst avoiding U-shaped obstacles. The distance can be shortened by reducing the temperature appropriately. A U-shaped obstacle is also set near the target. At this point, the temperature of the system is relatively low, and the obstacle is attempting to trap the robot. At this time, the temperature must increase to help the robot escape and then slowly cool down until it reaches the target.

Table 1 lists the simulation results in the aforementioned environments. The pathfinding time of the DA-APF is 30% less than that of the standard APF trapped in the local minimum, which shows that the DA-APF still maintains high efficiency in the face of U-shaped obstacle environments.

Table 1

Simulation results of the two algorithms (mean values from 50 runs for two algorithms).

	Standard APF	DA-APF
Environment One		
Time (s)	6.29	6.12
Path length	328 ^a	948
Environment Two		
Time (s)	7.06	2.22
Path length	279 ^a	502
Environment Three		
Time (s)	6.31	4.15
Path length	205 ^a	793

where

^aMeans not reaching the target.

4.3. Experiment two

This section uses the DA-APF to simulate the stochastic simple multi-objective environment. The simulation environment consists of 50 square obstacles with side lengths of 2 or 4, their positions are random, and the size is 100 dm × 100 dm. This environment can be regarded as a transition from a local trap environment to a global complex environment. It also aims to study the effects of some parameters in the algorithm on the final path found. The coordinates of the source and the target are (0, 0) and (100, 100), respectively. In each case, the DA-APF runs 10 times and the success rate and the average number of iterations of the pathfinding are recorded.

Fig. 11 shows that in similar obstacle environments, for a given $T_0 = 10000$, $\alpha = 0.98$, the final paths obtained under

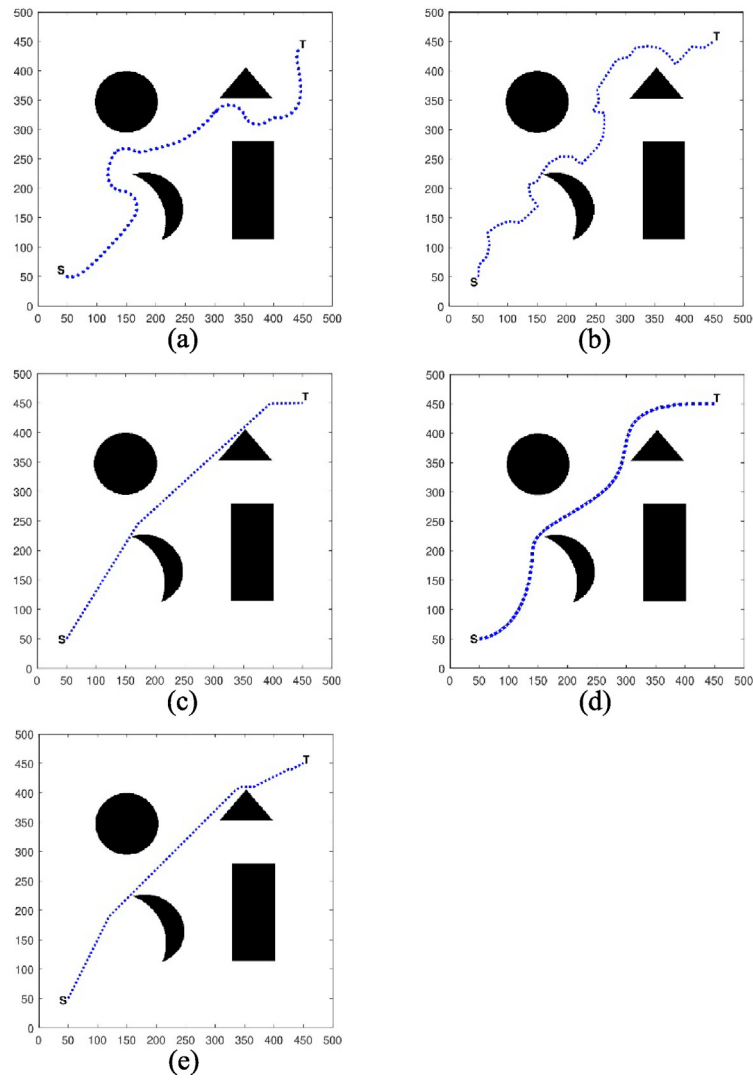


Fig. 12. Environment One (from (a) to (e): DA-APF, B-RRT, PRM, GA and IA*).

different step-sizes are similar. Therefore, the effects of the initial temperature T_0 and the cooling factor α are considered.

Table 2 shows the effects of different T_0 and α on path planning. The higher initial temperature improves the efficiency of the DA-APF algorithm with a pathfinding success rate close to 100%, which is also in line with the basic idea of the DA algorithm itself for high-temperature initialisation. According to the theory in Section 2, the reasonable value of cooling factor α is between 0.95 and 0.99. Thus, experiments are conducted according to this range. The results show that the DA-APF performs well at a higher cooling rate, but once the cooling rate is reduced, the algorithm performs poorly, and the pathfinding success rate even drops directly to 0. The reason for the analysis is that when α is less than the theoretical value, the temperature in the annealing process drops very quickly, which cannot meet the theoretical requirements of slow temperature decline and may even cause quenching, leading to poor algorithm performance. Meanwhile, because of $T_{i+1} = \frac{1}{\alpha}T_i$, a small α will lead to a fast-tempering process. According to the conclusions of Sections 4.2.2 and 4.2.3, the robot may deviate from the target or hit an obstacle to a large extent, resulting in a low pathfinding success rate. Therefore, the algorithm should be initialised at a sufficiently large temperature

Table 2

Simulation results of changing T_0 and α in DA-APF.

T_0	α	Average number of iterations	Success rate/%
10 000	0.98	610	100
	0.97	621	100
	0.96	624	100
	0.95	1339	80
	0.94	255	0
5000	0.98	1523	90
	0.97	1498	100
	0.96	1567	20
	0.95	1412	0
	0.94	174	0

first, whilst the cooling rate should be strictly controlled to ensure the stability of the annealing and tempering processes and improve the performance of the algorithm.

4.4. Experiment three

In this section, the DA-APF and other comparison algorithms are simulated in a complex environment. These environments

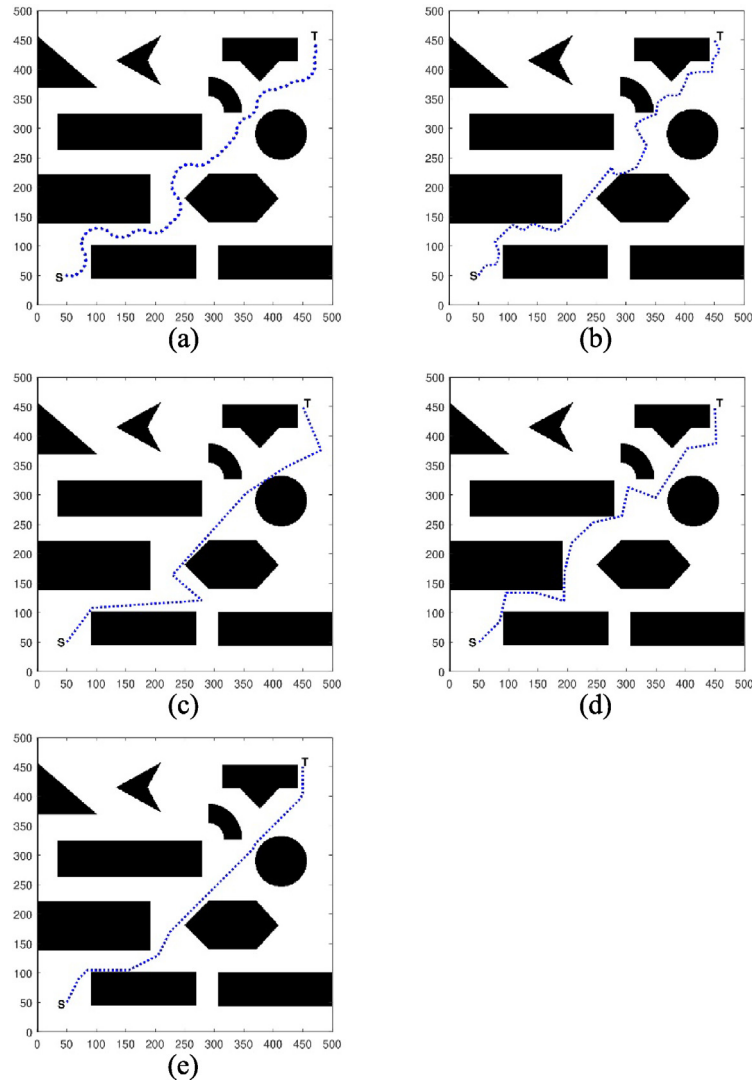


Fig. 13. Environment Two (from (a) to (e): DA-APF, B-RRT, PRM, GA and IA*).

include obstacles with convex or non-convex polygon, as well as open and partially closed scenes. The size of the environment is 500 dm $\times$ 500 dm. The coordinates of the source and the target are (50, 50) and (450, 450). According to the conclusion of Section 4.3, set $T_0 = 10\,000$, and $\alpha = 0.98$.

4.4.1. Environment One

Fig. 12 shows four convex and non-convex polygon obstacles, including a U-shaped obstacle, in this environment. The simulation results show that DA-APF and several other comparison algorithms can reach the target, and the success rate is close to 100%. According to the results in Table 3, GA takes the least time, 1.50 s, and IA* obtains the shortest path, 58.6 m. From the path trajectory, the path smoothness of B-RRT is significantly lower than those of the other algorithms. DA-APF does not exert its advantages in this environment, because U-shaped obstacles cause some trouble whilst other algorithms can ignore this type of obstacle.

4.4.2. Environment Two

This environment increases the number, variety and size of obstacles, which can function as a transitional scene. Fig. 13 shows that the paths obtained by various algorithms are roughly the same. In terms of the smoothness of these paths, DA-APF and IA*

Table 3

Simulation results of Environment One (mean values from 50 runs for all algorithms).

	Time	Path length	Success rate/%
DA-APF	1.90	713	100
B-RRT	2.89	766	100
PRM	4.56	599	96
GA	1.50	794	98
IA*	5.98	586	100

perform remarkably well. According to the results in Table 4, the success rate of DA-APF and B-RRT reached 100%, which has great advantages over other algorithms. Similar to Environment One, GA takes the least time, 1.54 s, and IA* obtains the shortest path, 68.1 m. The paths obtained by DA-APF maintain a certain safe distance from each obstacle due to the setting of the threshold distance ρ_0 . Although the other algorithms have also completed obstacle avoidance, many instances of imminent collision need to be improved in actual environments.

4.4.3. Environment Three

Fig. 14 shows that this environment is set as a closed scene, which is more in line with the modelling of path planning in actual environments. In this environment, the paths obtained by

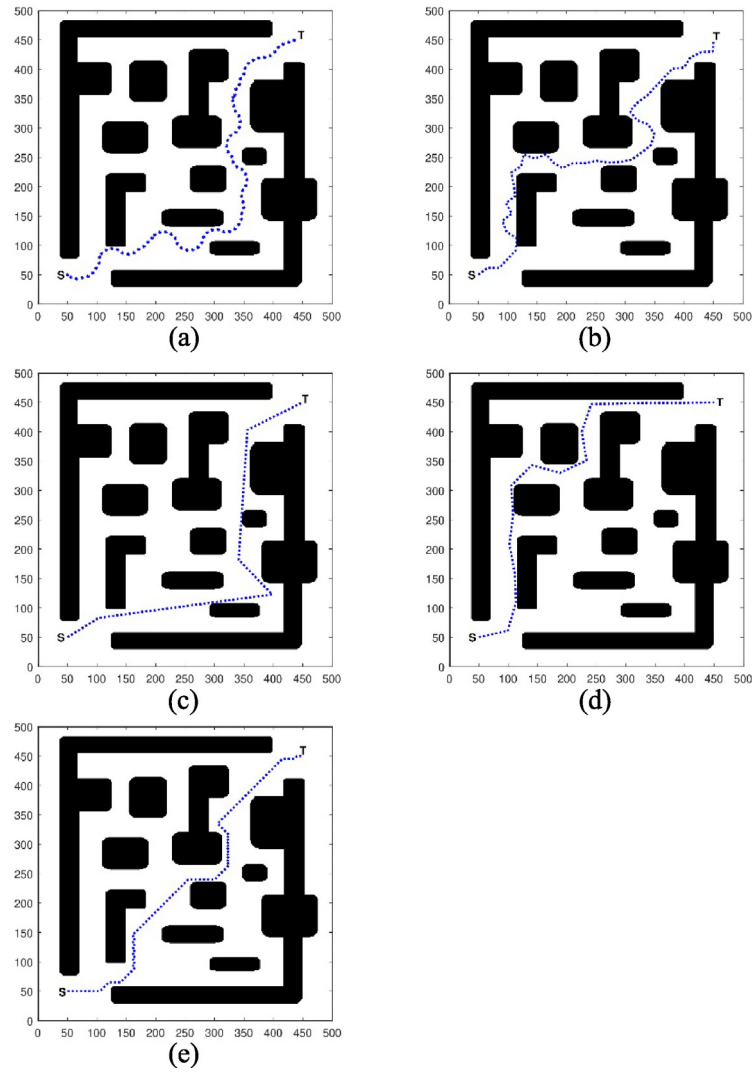


Fig. 14. Environment Three (from (a) to (e): DA-APF, B-RRT, PRM, GA and IA*).

Table 4

Simulation results of Environment Two (mean values from 50 runs for all algorithms).

	Time	Path length	Success rate/%
DA-APF	1.75	710	100
B-RRT	2.58	714	100
PRM	1.88	733	88
GA	1.54	754	90
IA*	4.56	681	94

various algorithms are different, and the path trajectories of DA-APF and PRM are relatively simple. Similarly, the smoothness of the path obtained by DA-APF is better than the other algorithms, which making it suitable for more realistic scenes. The detailed data are shown in Table 5. DA-APF takes the least time, 1.90 s, and IA* obtains the shortest path, 75.7 m. Meanwhile, the success rate of DA-APF and B-RRT reached 100%. Compared with Environment One, no obvious U-shaped obstacle can be observed, and the proposed DA-APF appears to obtain satisfactory results.

4.4.4. Environment Four

Fig. 15 intercepts from a real maze map, which is almost closed and regular. The five algorithms provide very similar paths, where the PRM algorithm gives the intuitively optimal path. As in

Table 5

Simulation results of Environment Three (mean values from 50 runs for all algorithms).

	Time	Path length	Success rate/%
DA-APF	1.90	770	100
B-RRT	2.80	771	100
PRM	2.61	778	96
GA	2.13	779	94
IA*	2.98	757	90

the previous experiments, DA-APF does a better job than several other algorithms in handling the point of safe distance from the obstacles. Table 6 shows that IA* gets the best average path length and DA-APF has better path finding time and excellent path quality. Comparing with all previous simulations, we can conclude that DA-APF has the optimal pathfinding time and the shortest path in the environment without U-shaped obstacles.

To show the performance of various algorithms more intuitively in the above environments, a comprehensive index I is proposed:

$$I_i = \frac{\text{Time}_i \times 0.25 + \text{Path length}_i \times 0.25 - \text{Success rate}_i \times 0.5}{\text{Time}_{\text{DA-APF}} \times 0.25 + \text{Path length}_{\text{DA-APF}} \times 0.25 - \text{Success rate}_{\text{DA-APF}} \times 0.5} \quad (19)$$

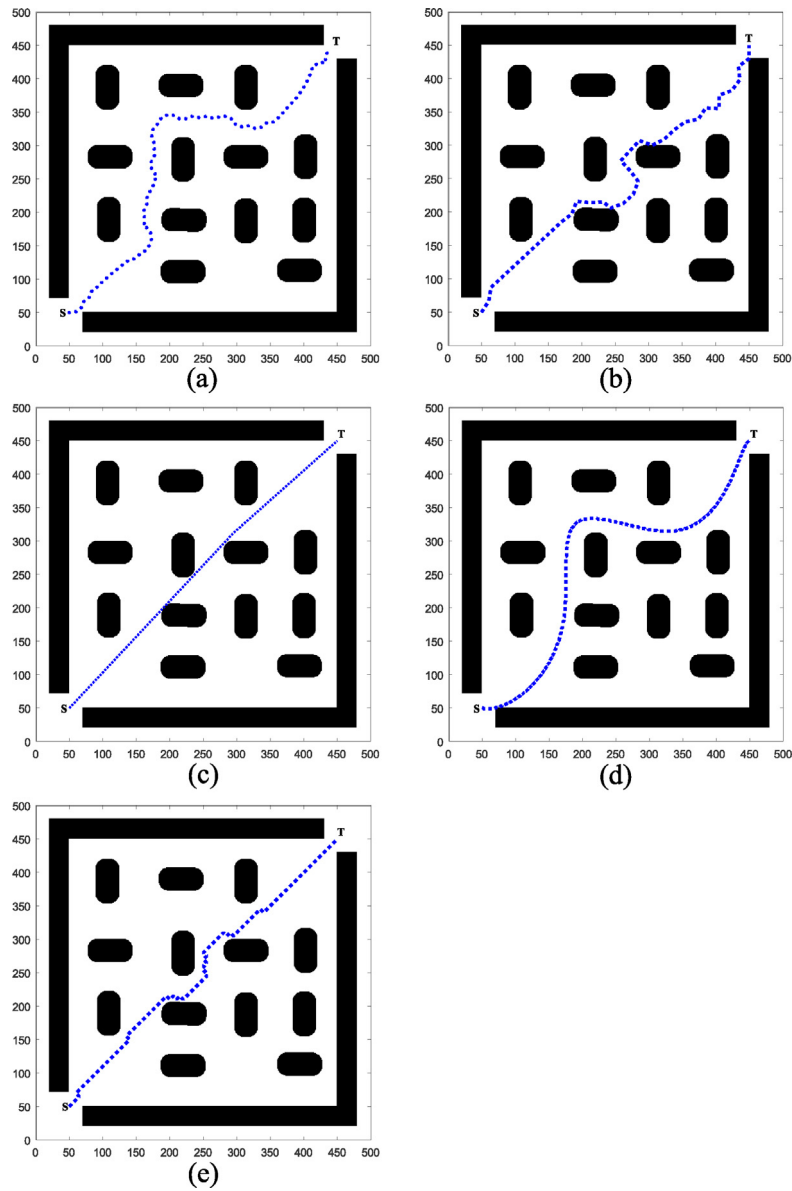


Fig. 15. Environment Four (from (a) to (e): DA-APF, B-RRT, PRM, GA and IA*).

Table 6

Simulation results of Environment Four (mean values from 50 runs for all algorithms).

	Time	Path length	Success rate/%
DA-APF	1.49	680	100
B-RRT	2.38	727	100
PRM	5.69	778	98
GA	4.48	762	80
IA*	3.96	674	84

Table 7

Comprehensive indices of five algorithms.

	Environment One	Environment Two	Environment Three	Environment Four
DA-APF	1	1	1	1
B-RRT	1.10	1.01	1.01	1.10
PRM	0.80	1.09	1.03	1.22
GA	1.16	1.12	1.04	1.26
IA*	0.76	0.97	1.01	1.06

The comprehensive indices of the five algorithms are calculated by Eq. (19), as shown in Table 7.

Taking DA-APF as the benchmark, if the index I is lower than 1, then an algorithm is better than DA-APF, otherwise, the algorithm is relatively poor. Therefore, PRM and IA* perform well in Environment One. In Environment Two, DA-APF is slightly inferior to IA* and better than the rest of the algorithms. In Environment Three, DA-APF obtains the best results. Although IA*

can always find the shortest path, it has obvious disadvantages in pathfinding time and success rate. The conclusion obtained in Environment Four is similar to that in Environment Three, where DA-APF always balances pathfinding efficiency and path quality. Because the path obtained by DA-APF is relatively smooth and has a safe distance guarantee, in general, DA-APF can solve the path planning problem in the four environments above and has excellent comprehensive performance.

5. Conclusions

This paper proposes a DA-APF algorithm based on a deterministic annealing strategy for robot path planning in static obstacle environments. Compared with the standard APF, DA-APF introduces a temperature parameter into the standard potential field function to increase the potential effect of the obstacles on the robot. It also simulates the annealing and tempering processes to help the robot escape from the local minima by using the gradient descent potential guided strategy to drive the robot to the target. In addition, by constructing a Lagrangian-type free energy potential function that corresponds to the APF system, the global minimum is addressed by taking the idea of deterministic annealing, and then the global minimum of the improved potential field function is obtained to optimise the selection of the initial path. The simulation results show that DA-APF can solve the path planning in U-shaped obstacle environments. Furthermore, this method exhibited the best performance under high initial temperatures and large cooling factors, with a pathfinding success rate close to 100%. Compared with algorithms such as B-RRT, DA-APF can obtain optimal or near-optimal paths in various complex environments, which have good smoothness and safe distances without oscillations. Given that this paper focuses on simulating static environments, in our future work, the real-time path planning for dynamic obstacles will be explored further.

CRediT authorship contribution statement

Zhengtian Wu: Formal analysis, Funding acquisition, Methodology, Software, Investigation, Writing – review & editing. **Jinyu Dai:** Data curation, Software, Writing – original draft. **Baoping Jiang:** Formal analysis, Investigation, Resources, Writing – original draft. **Hamid Reza Karimi:** Conceptualization, Investigation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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